1. Clinical Question & Core Hypothesis

Clinical Question

After radical prostatectomy some patients show 68Ga-PSMA-11–positive lesions despite low serum PSA, and these discordant imaging/biochemical patterns variably predict early recurrence — why does PSMA-PET detect progressive disease that PSA alone misses, and which biology underlies the higher risk of progression? 2) Tier 1 path used: the single provided path linking radical prostatectomy → PCA → staging by 68Ga-PSMA-11 ↔ biochemical context (serum PSA / PSA velocity) → recurrence/progression → FFS; I built on the mechanistic inference that PSMA PET signal (PSMA/FOLH1 biology) can be decoupled from PSA (KLK3) secretion. 3) Supporting evidence (Tier 2 equivalence): PSA velocity and trial-informed therapy decision nodes (docetaxel, STAMPEDE, prostate RT) support that dynamic PSA kinetics plus imaging inform therapy timing and influence progression-free outcomes. 4) Why PFI: the path centers on recurrence/progression and failure-free survival; PFI (progression-free interval) directly captures the timing of biochemical/imaging-detected recurrence and is the most relevant endpoint for evaluating whether integrated imaging/PSA dynamics change clinical course. 5) Deliberate exclusions: I excluded adding additional molecular markers or broader clinical variables to keep the hypothesis sharply mechanistic (max 2 anchor genes). I did not include STAMPEDE as a core anchor because it is a trial context rather than a mechanistic biomarker; it is retained as a secondary modifier. I also excluded speculative genes not present in the path?

Working Hypothesis

Primary hypothesis: In post-prostatectomy patients with biochemical recurrence, 68Ga-PSMA-11–avid lesions detected at low serum PSA levels reflect a biology characterized by high FOLH1 (PSMA) expression with relatively low KLK3 (PSA) secretion. When combined with PSA velocity, this PSMA-imaging signal provides superior prediction of shorter progression-free interval (PFI) than PSA metrics alone. Secondary hypotheses: (a) A predictive model that integrates 68Ga-PSMA-11 PET positivity and PSA velocity will stratify patients into risk groups with distinct PFI. (b) Early salvage therapy decisions guided by PSMA-PET findings (e.g., targeted salvage radiotherapy to PSMA-avid sites and/or earlier systemic therapy such as docetaxel) will improve PFI compared with PSA-guided timing. (c) Findings are robust across clinically relevant variations in PSMA-PET interpretation, anchored specifically to FOLH1 and KLK3 biology.

Element	Value
Imaging feature	68Ga-PSMA-11
Anchor gene(s)	FOLH1, KLK3
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

FOLH1: FOLH1 biology) can be decoupled from PSA (KLK3) secretion.

KLK3: KLK3) secretion.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PSA_velocity	Supporting modifier from knowledge graph
Secondary	docetaxel	Supporting modifier from knowledge graph
Secondary	prostate_RT	Supporting modifier from knowledge graph
Secondary	FFS	Supporting modifier from knowledge graph
Secondary	STAMPEDE	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: expression drives 68Ga-PSMA-11 PET positivity, enabling detection of recurrent/metastatic disease that may be occult to PSA-based surveillance if KLK3 secretion is disproportionately low. PSA velocity is an established prognostic indicator for progression in recurrent prostate cancer.

Prediction 2: Retrospective cohort of post-prostatectomy patients with biochemical recurrence who underwent 68Ga-PSMA-11 PET, with data on PSA level, PSA velocity, imaging results, treatments (salvage RT, docetaxel), and follow-up for PFI.

Prediction 3: Statistical modeling: Cox proportional hazards analyses comparing models using PSA alone vs. PSA plus PSMA-PET status, including interaction with PSA velocity.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PSA_velocity: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- docetaxel: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.